

Taller 1, puntos 7 a 9: Adaline y compuertas

Entradas en [0,1], salida deseada en [0,1], criterio de parada por error cuadrático

```
clear; close all; clc
```

Parametros del modelo

```
alpha      = 0.05;  
umbral     = 0.5;      % escalon para clasificar la salida lineal  
salida     = [0 1];  
error_min  = 1e-3;  
max_epocas = 1000;  
semilla    = 1;
```

OR de 2 entradas

```
[X, d] = compuerta('OR', 2, salida);  
[w, b, epocas, EC] = adaline(X, d, alpha, error_min, max_epocas, semilla);  
epocas, EC_final = EC(end)
```

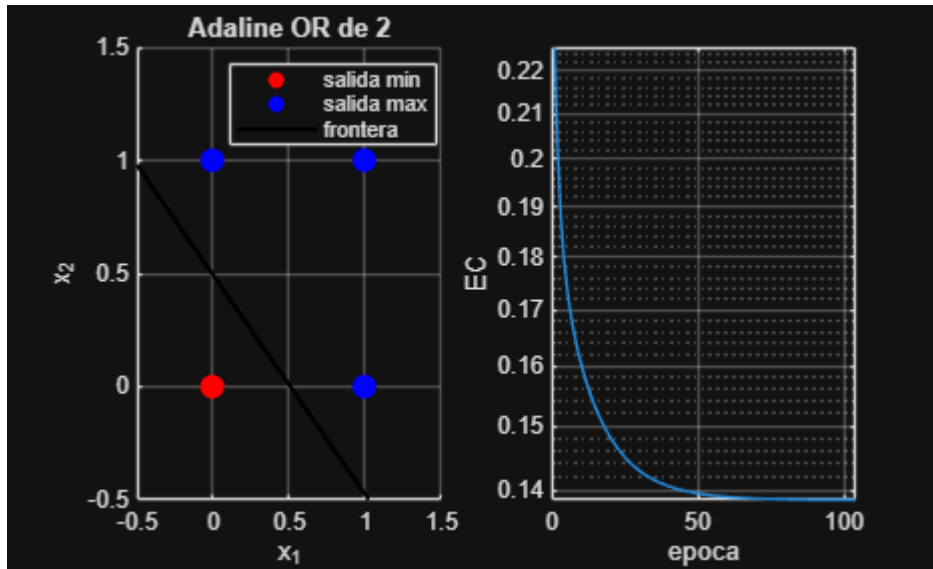
```
epocas =  
104  
EC_final =  
0.1385
```

```
table(X(:,1), X(:,2), d, X*w' + b, predecir(X, w, b, umbral, salida),  
      'VariableNames', {'x1','x2','deseada','lineal','clasificada'})
```

```
ans = 4x5 table
```

	x1	x2	deseada	lineal	clasificada
1	0	0	0	0.2574	0
2	0	1	1	0.7495	1
3	1	0	1	0.7350	1
4	1	1	1	1.2271	1

```
figure  
subplot(1,2,1); graficar_frontera(X, d, w, b, umbral, 'Adaline OR de 2');  
subplot(1,2,2); semilogy(EC); grid on; xlabel('epoca'); ylabel('EC')
```



Reporte 1: todas las compuertas, incluida la XOR

```
casos = {'AND',2; 'AND',3; 'AND',4; 'OR',2; 'OR',3; 'OR',4; 'XOR',2};
R1 = table();
for c = 1:7
    [X, d] = compuerta(casos{c,1}, casos{c,2}, salida);
    [w, b, ep, EC] = adaline(X, d, alpha, error_min, max_epocas, semilla);
    acc = 100*mean(prededir(X, w, b, umbral, salida) == d);
    R1 = [R1; table(casos{c,1}, casos{c,2}, ep, EC(end), acc, 'VariableNames',
{'Compuerta','Entradas','Epocas','EC_final','Exactitud'})];
end
R1
```

R1 = 7x5 table

	Compuerta	Entradas	Epocas	EC_final	Exactitud
1	'AND'	2	67	0.1385	100
2	'AND'	3	115	0.2953	100
3	'AND'	4	59	0.4292	93.7500
4	'OR'	2	104	0.1385	100
5	'OR'	3	53	0.2727	87.5000
6	'OR'	4	86	0.3697	93.7500
7	'XOR'	2	128	0.5540	25

Reporte 2: efecto de alpha

```
alphas = [0.001 0.01 0.05 0.1 0.2 0.5 1];
R2 = table();
for c = 1:6
    [X, d] = compuerta(casos{c,1}, casos{c,2}, salida);
```

```

for a = alphas
    [w, b, ep, EC] = adaline(X, d, a, error_min, max_epocas, semilla);
    acc = 100*mean(predecir(X, w, b, umbral, salida) == d);
    R2 = [R2; table(casos(c,1), casos{c,2}, a, ep, EC(end), acc,
'VariableNames', {'Compuerta', 'Entradas', 'Alpha', 'Epocas', 'EC_final', 'Exactitud'})]];
end
end
R2

```

R2 = 42×6 table

	Compuerta	Entradas	Alpha	Epocas	EC_final	Exactitud
1	'AND'	2	1.0000e-03	1000	0.1290	100
2	'AND'	2	0.0100	327	0.1276	100
3	'AND'	2	0.0500	67	0.1385	100
4	'AND'	2	0.1000	27	0.1541	100
5	'AND'	2	0.2000	47	0.1953	100
6	'AND'	2	0.5000	20	0.5000	75
7	'AND'	2	1	1000	4.9943e+05	50
8	'AND'	3	1.0000e-03	1000	0.2590	87.5000
9	'AND'	3	0.0100	282	0.2583	100
10	'AND'	3	0.0500	115	0.2953	100
11	'AND'	3	0.1000	71	0.3516	100
12	'AND'	3	0.2000	43	0.5068	87.5000
13	'AND'	3	0.5000	27	1.2444	62.5000
14	'AND'	3	1	504	Inf	87.5000

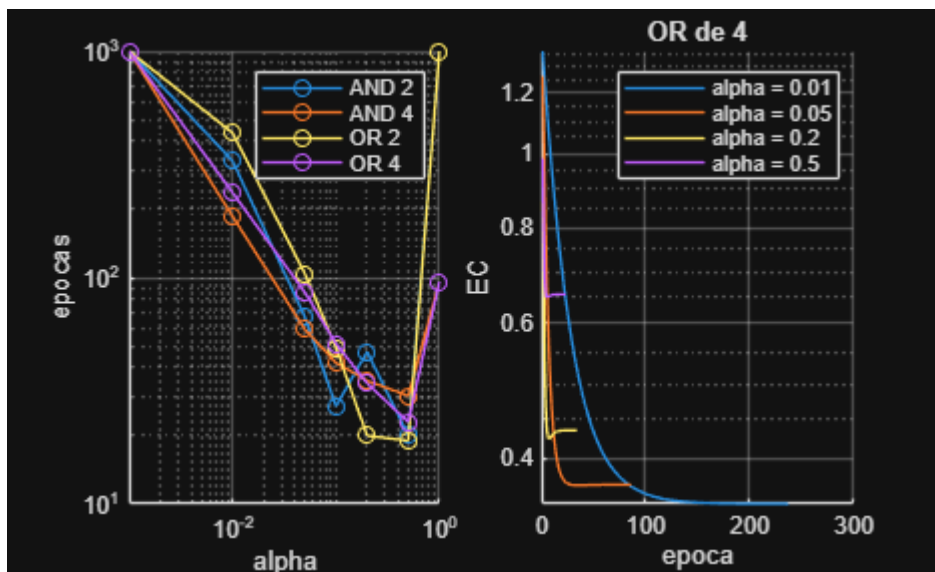
⋮

```

figure
subplot(1,2,1); hold on
for c = [1 3 4 6]
    m = strcmp(R2.Compuerta, casos{c,1}) & R2.Entradas == casos{c,2};
    plot(R2.Alpha(m), R2.Epocas(m), '-o', 'DisplayName', sprintf('%s %d',
casos{c,1}, casos{c,2}));
end
set(gca, 'XScale', 'log', 'YScale', 'log'); grid on; legend; xlabel('alpha');
ylabel('epocas')
subplot(1,2,2); hold on
[X, d] = compuerta('OR', 4, salida);
for a = [0.01 0.05 0.2 0.5]
    [~, ~, ~, EC] = adaline(X, d, a, error_min, max_epocas, semilla);
    semilogy(EC, 'DisplayName', sprintf('alpha = %g', a));
end

```

```
set(gca, 'YScale', 'log'); grid on; xlim([0 300]); legend; xlabel('epoca');  
ylabel('EC'); title('OR de 4')
```



Reporte 3: tolerancia de parada y umbral (AND de 2)

Con una tabla de verdad el EC nunca llega a $1e-3$, el que para es el cambio entre epocas

```
[X, d] = compuerta('AND', 2, salida);  
R3 = table();  
for tol = [1e-2 1e-3 1e-4 1e-5 1e-6]  
    for u = [0.3 0.5 0.7]  
        [w, b, ep, EC] = adaline(X, d, alpha, error_min, max_epocas, semilla, tol);  
        acc = 100*mean(prededir(X, w, b, u, salida) == d);  
        R3 = [R3; table(tol, u, ep, EC(end), acc, 'VariableNames',  
{'Tol_cambio', 'Umbral', 'Epocas', 'EC_final', 'Exactitud'})];  
    end  
end  
R3
```

R3 = 15x5 table

	Tol_cambio	Umbral	Epocas	EC_final	Exactitud
1	0.0100	0.3000	6	0.1622	75
2	0.0100	0.5000	6	0.1622	100
3	0.0100	0.7000	6	0.1622	100
4	1.0000e-03	0.3000	13	0.1464	75
5	1.0000e-03	0.5000	13	0.1464	100
6	1.0000e-03	0.7000	13	0.1464	100
7	1.0000e-04	0.3000	34	0.1392	100
8	1.0000e-04	0.5000	34	0.1392	100

	Tol_cambio	Umbral	Epocas	EC_final	Exactitud
9	1.0000e-04	0.7000	34	0.1392	100
10	1.0000e-05	0.3000	54	0.1385	100
11	1.0000e-05	0.5000	54	0.1385	100
12	1.0000e-05	0.7000	54	0.1385	100
13	1.0000e-06	0.3000	67	0.1385	100
14	1.0000e-06	0.5000	67	0.1385	100

⋮

Perceptron contra Adaline en las mismas compuertas

```
R4 = table();
for c = 1:7
    [X, d] = compuerta(casos{c,1}, casos{c,2}, [0 1]);
    [wp, bp, ep_p] = perceptron(X, d, 3, 0.05, 0.5, [0 1], 1000, 1);
    [wa, ba, ep_a] = adaline(X, d, 0.05, error_min, max_epocas, 1);
    R4 = [R4; table(casos(c,1), casos{c,2}, ep_p, 100*mean(prededir(X,wp,bp,0.5,[0
1]) == d), ...
                    ep_a, 100*mean(prededir(X,wa,ba,0.5,[0 1]) == d), ...
                    'VariableNames',
                    {'Compuerta', 'Entradas', 'Epocas_perc', 'Exact_perc', 'Epocas_ada', 'Exact_ada'})]];
end
R4
```

R4 = 7×6 table

	Compuerta	Entradas	Epocas_perc	Exact_perc	Epocas_ada	Exact_ada
1	'AND'	2	4	100	67	100
2	'AND'	3	7	100	115	100
3	'AND'	4	6	100	59	93.7500
4	'OR'	2	2	100	104	100
5	'OR'	3	3	100	53	87.5000
6	'OR'	4	5	100	86	93.7500
7	'XOR'	2	1000	50	128	25